Chapter 7 · Solutions

P33 · Grain before aggregation

Hint. Count each fact row's join partners.

The grain is one sale line. The star has one fact and three dimension tables: 4. Splitting category information into its own dimension relation gives 5 tables and a snowflaked product dimension.

The true sale sum is 100+50=150. A has two tag partners and B has one, so the joined amounts are 100,100,50, summing to 250. The join changes the grain to sale-line×tag. To sum sales, retain one contribution per sale line or use an existence filter when tags only determine eligibility.

P34 · Describe the actual change

Country→State is drill-down. City→State is roll-up. Fixing one Country value is a slice. Restricting multiple dimensions is a dice. Swapping axes is a pivot.

A larger output count does not prove drill-down. Extending a report from one month to two months can add rows while keeping its grouping levels unchanged. The levels, not the count alone, decide drill-down or roll-up.

P35 · Three quantities that look similar

Hint. Count observed groups within each grouping set.

Grouping set	Original groups	After adding (B,US,40)
Product,Country	3	4
Product	2	2
Country	2	2
no dimensions / grand total	1	1
Total output rows	8	9

There are 2^2=4 grouping sets in both cases. The first dataset is missing one detail combination, so the complete-product formula 3×3=9 does not apply to it.

For independent hierarchies, choosing one level per dimension gives 4×3×4=48 cuboids. Those counts already include ALL; adding it again would overcount.

P36 · Carry sufficient summaries

Hint. Recover totals, then divide by the total count.

The group sums are 2×100=200 and 8×50=400. Overall mean=(200+400)/(2+8)=60. Carry sum and count. AVG is algebraic; averaging the two subgroup averages without weights would give the wrong value 75.

The customer union is {1,2,3}, so the distinct count is 3. Adding subgroup counts gives 4 and counts customer 2 twice. Counts 2,2 could also describe disjoint sets with union size 4, proving that counts alone cannot determine the answer.

Balances can be summed across accounts at the same date. Adding balances across dates usually counts the same stock repeatedly and does not describe a balance at any date. Balance is semi-additive across those dimensions. This differs from the distributive/algebraic/holistic classification of computation.